

# CDS Compute Cluster

## Showcase Presentation

**Cornell Data Science**

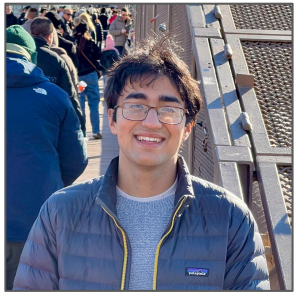
Fall 2025

# Agenda

1. Introduction
2. Architecture
3. LLM Hosting
4. Website and Project Hosting
5. Reflections & Next Steps

# Introduction

# Meet the team!



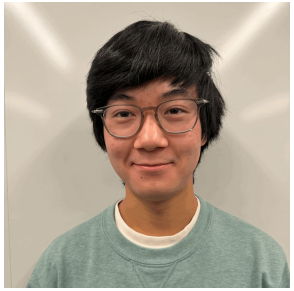
**Mukund Gaur**
Tech Lead | CS ‘28



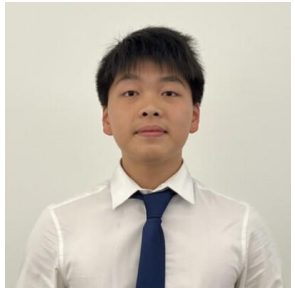
**Rahul Basak**
Project Manager | CS ‘28



**Rohit Vakkalagadda**
CS ‘28



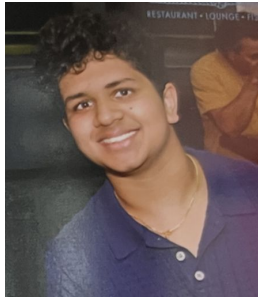
**Henry Ji**
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**Naijei Jiang**
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**Aden Zhao**
CS ‘28



**Srikar Karra**
CS ‘28

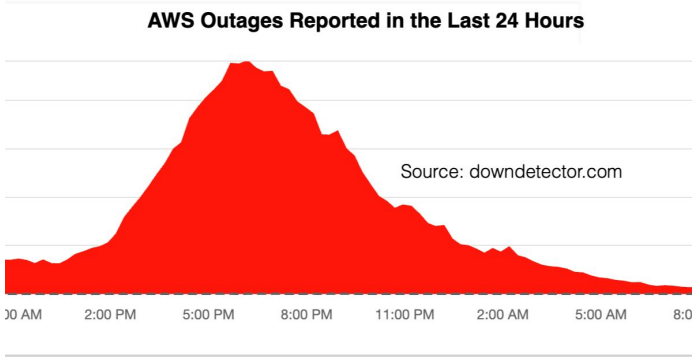


**Vedanta Jain**
CS ‘28

# Architecture

# Motivation

- In-house replacement for AWS
  - Save money on cloud computing, hosting ML models, training
  - Reduce dependency on external cloud providers and APIs
- Local LLM hosting
- Proof-of-concept of distributed computing
  - Can be customized/expanded as needed
- Build out the infrastructure for future projects!
  - “Here’s the compute, go build it.”



# Hardware & Location

Location:

Experiential Learning Lab (Upson Hall)







Hardware

- Pi 4
- Pi 5
- Gaming PC
- Dell Desktop
- Alienware Aurora R7
- Laptop Compute Nodes
- Network Switch

# Networking

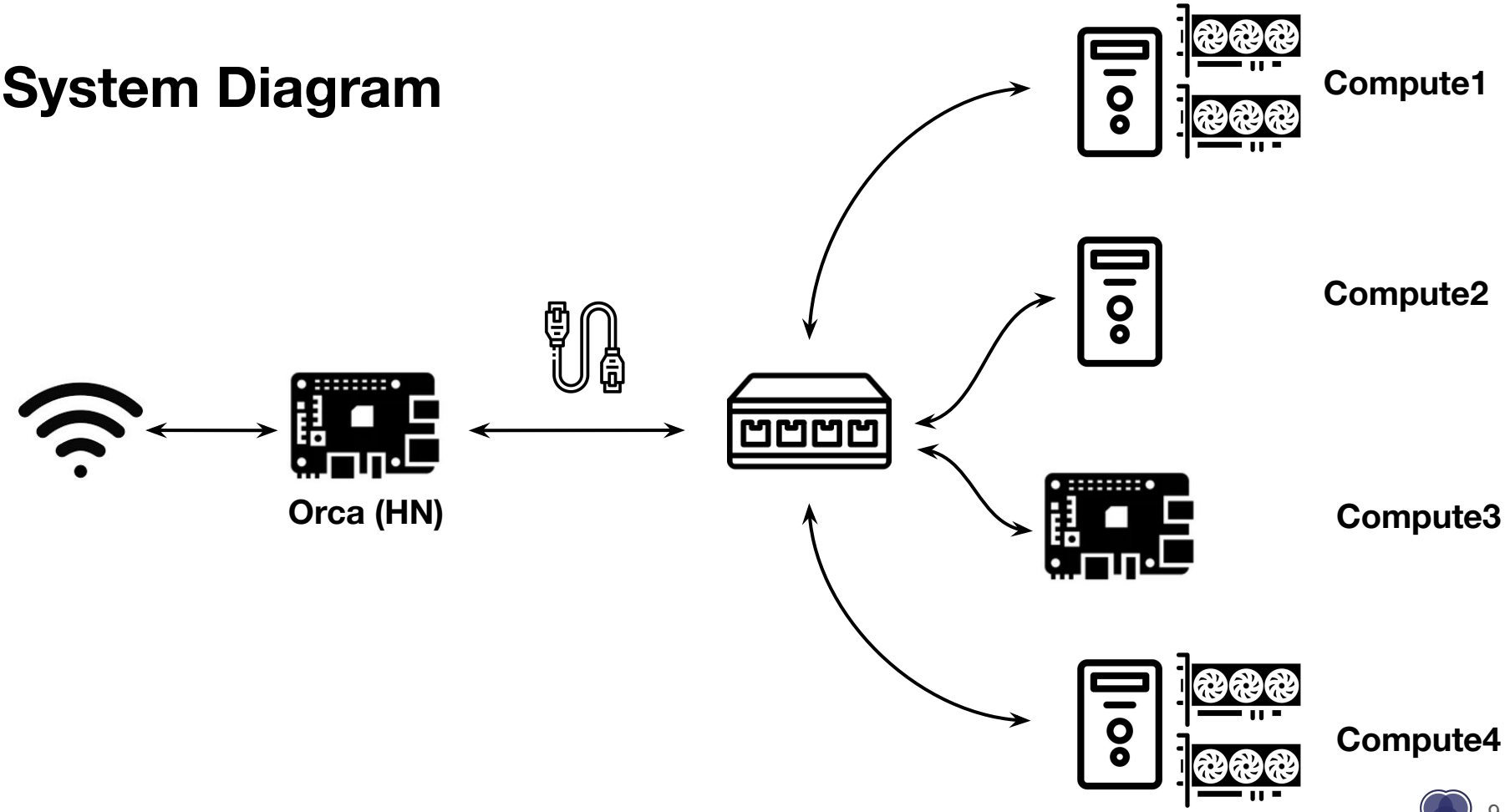
## Head Node (Pi4)

- Connected to the internet via RedRover
- Serves as login node and interface point between users and the cluster
- Forwards internet traffic to compute nodes if needed

## Private Ethernet Subnet

- Connects compute nodes to each other and to the head node over a network switch, which allows the nodes to have their own LAN
- Network switch forwards packets over the LAN
- Each node has a local IP address on the private ethernet subnet
    - Head node also has a public IP in addition to this

# System Diagram



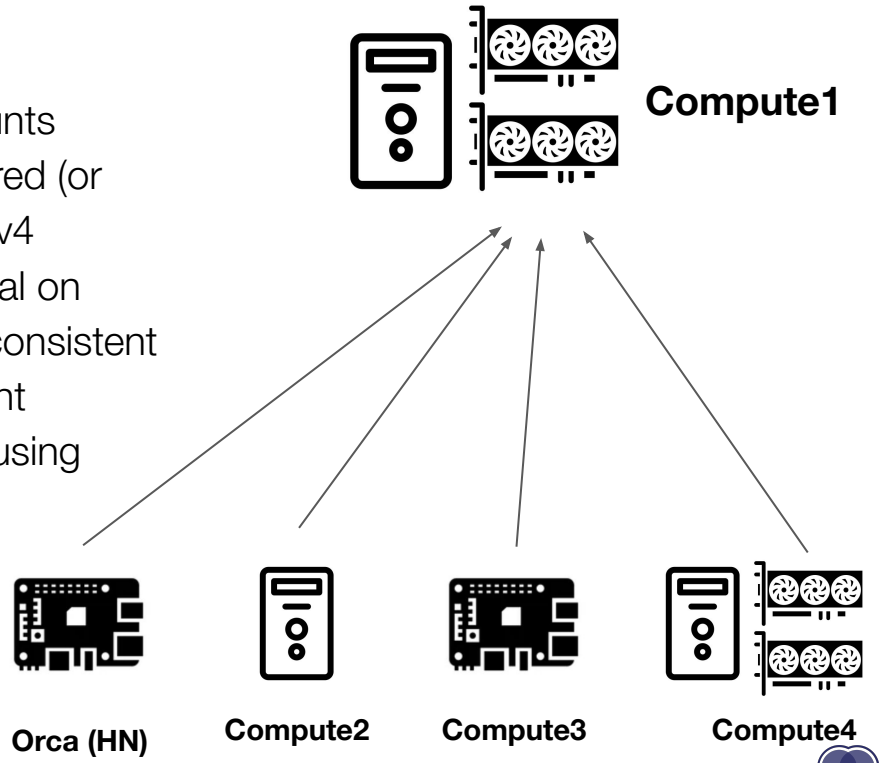
# File Storage

## Architecture

- Compute1 runs the NFS Kernel Server
- All shared data is stored directly on Compute1 with safe concurrent access
- Orca + all compute nodes install NFS Common Client and access that storage

## Mounting

- Each node mounts compute1:/shared (or /home) via NFSv4
- Appears identical on every node → consistent user environment
- Auto-mounted using /etc/fstab



# Job Scheduling

## Munge

- Provides secure authentication between all cluster nodes
- Ensures Slurm daemons trust requests from users and other nodes

## Chrony

- Synchronizes system clocks across the cluster
- Prevents time drift that would break Slurm communication and job scheduling

## Slurm

- Schedules and runs jobs across compute nodes
- Allocates CPUs, GPUs, and memory while managing job queues and priorities

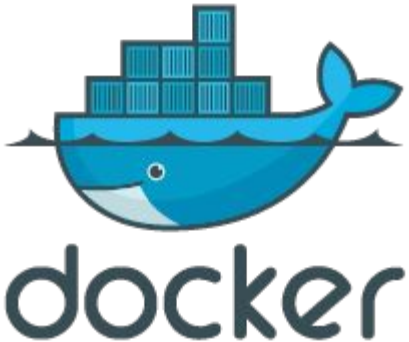





# Docker setup

Docker: Used to manage environments across nodes

- Head node serves as registry for images - create and push images
- Compute nodes can pull from the registry by accessing a port served by the head node - allows for consistent images across nodes
- Especially when creating unique images for ARM vs x64 computer architectures





# Architecture Walkthrough

# LLM Hosting

# vLLM Setup

## What is vLLM?

- High-performance inference engine for Large Language Models (LLMs)
- Supports tensor parallelism across multiple GPUs

## Cluster Setup

- Serving Llama-2-7b-chat-hf on compute1 with two RTX 2080 Ti GPUs
- Slurm job scheduler manages GPU allocation and job queueing
- Server runs as persistent Slurm job, custom shell scripts automate job submission and result retrieval

## What is the point?

- No per-token charges, unlimited inference on private hardware
- All data stays internal; no external API calls
- Full control over model selection, parameters, configurations



# vLLM Demo

# Llama.cpp

## What is it?

- High-performance inference engine written in C/C++
- No heavy overhead from Python/PyTorch dependencies



## Cluster Setup

Currently hosting Llama-3.1-8B on the two 2080s in compute1

- **`llamacppchat`** opens a chat window with Llama
- **`llamacppprompt`** allows the user to send one prompt
- **`llamacppwarm`** will preload in the model to reduce overhead

## What is the point?

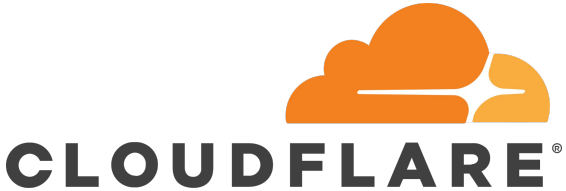
- Local LLM for anyone on the team - will support any GGUF model



# Llama.cpp Demo

# Website & Project Hosting

# CDS Website & Cloudflare Tunnels




## Cloudflare Tunnels

- `cloudflared` agent runs on head node and establishes an outbound connection to Cloudflare edge servers
- Any traffic to public hostname is forwarded over the already open connection to your machine
  - Benefits: Safer, no need to expose a port and IP, automatic routing, agent is lightweight
- Free service!

## CDS Website

- Web server runs locally on a compute node and accessible to the head node on the private subnet (e.g. 10.0.0.4:3000), cloudflare tunnel runs on head node

# Health Dashboard

## Backend Scripts

- Python scripts directly parse Linux virtual filesystems (/proc, /sys) to harvest raw hardware telemetry (thermal, RAM, I/O) without external dependencies.
- Harvests CPU temperature, active processes, RAM, and Disk stats every 3 seconds.
- All data are written to a shared JSON files which are then displayed on the frontend

## Front End

- Web server runs locally on a compute node and accessible to the head node on the private subnet (e.g. 10.0.0.4:3000), cloudflare tunnel runs on head node
- Located at orca.cornelldata.science



# Websites Demo

# MathSearch: Original Pipeline

S3

- AWS object storage service, stores objects in buckets — designed to be able to store and retrieve any amount of data from the web
- **MathSearch:** Uses S3 to store pdf files and images

SQS

- AWS fully managed message queueing system
- **MathSearch:** Uses SQS to transmit requests from the front end to the backend asynchronously

SageMaker

- AWS cloud-based service to allow developers to train and deploy ML models at scale
- **MathSearch:** Uses SageMaker to host YOLO model

# ClusterMathSearch: New Pipeline



MathSearch

## Local File Storage

- `/input` and `/output` directories on `compute1`
- **ClusterMathSearch** uses these directories to store pdfs and images locally

## File Monitor

- Monitors the `/input` directory for new job payloads every 2 seconds
- Automatically triggers Lambda functions to process incoming tasks
- **ClusterMathSearch** runs a monitor script as middleware between frontend and backend

## Model Hosting

- Replaced SageMaker inference with local server calls
- **ClusterMathSearch** hosts the YOLO model on `compute1`

# ClusterMathSearch: Front End

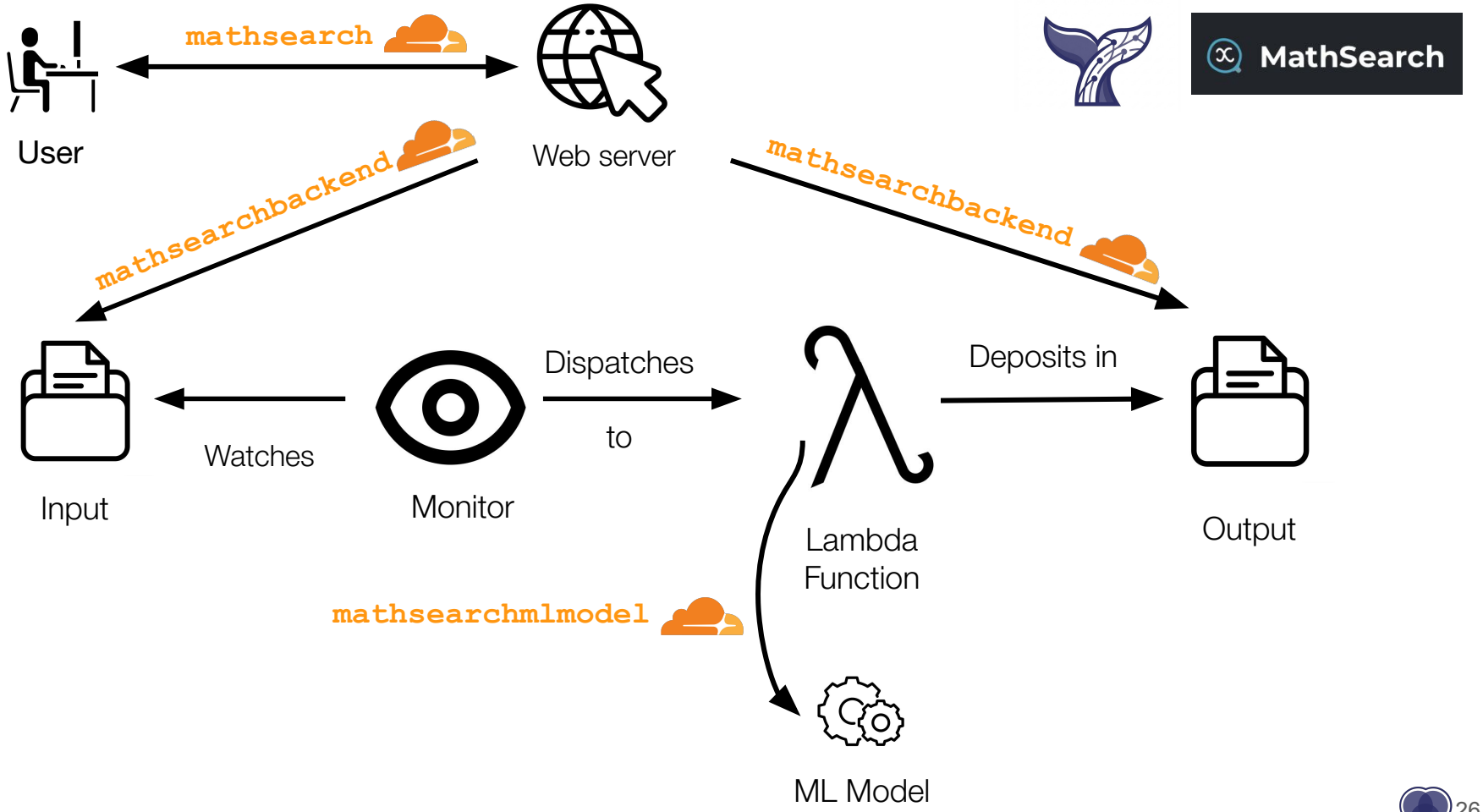


MathSearch

Cloudflare Tunnels

- `mathsearchmlmodel`: Exposes the server with inference API endpoints
- `mathsearchbackend`: Exposes our locally hosted file storage solution (allows upload of pdfs and querying of results)
- `mathsearch`: Exposes front end

All S3 queries replaced with calls to mathsearchbackend, all SageMaker calls replaced with calls to `mathsearchmlmodel`


mathsearch
User
Web server
MathSearch
mathsearchbackend
mathsearchbackend
Input
Watches
Monitor
Dispatches
to
Lambda
Function
Deposits in
Output
mathsearchmlmodel
ML Model



# MathSearch Demo

# Reflection & Takeaways

# Original Objectives & Results

Primary Objective: Fully Functional Cluster

- Result: Built out cluster with 4 compute nodes, a head node, job scheduler, and deployed a locally hosted LLM for CDS members to use

Part 5: Beyond the Infrastructure

- Docker
- Health dashboard
- Website hosting
- vLLM and Llama.cpp
- MathSearch

# Things We Learned

- How Linux works, how operating systems work
- Basics of the internet and computer networking
- How computer architectures affect programs (e.g. ARM vs AMD)
- Computer hardware (SSDs, DDR2 vs DDR3 vs DDR4 vs laptop RAM), hardware troubleshooting)
- Basics of file systems, network file storage, mounting directories, repairing file systems
- AWS services (S3, SQS, Lamda)
- Frontend and backend hosting
- Distributed training and inference
- Adapting to very different architectures

Tools: Slurm, vLLM, Llama.cpp, Docker, Python, JavaScript, C++, CSS, HTML, Cloudflare, CUDA, Linux, AWS, LLMs

# Next Steps?

# Thanks for listening!
# Questions?